Discrete Random Variables

Questions 1–40

Question 1. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3	4	5
$\mathrm{P}(X = x)$	0.11	0.13	0.29	0.30	0.17

Find $\mathrm{P}(2 \leq X < 5)$

Question 2. A probability distribution is given by

$$\mathrm{P}(X = x) = \frac{x}{100} + \frac{1}{5}$$

for $x = 2, 4, 6, 8$. Draw a table showing the probability distribution of X.

Question 3. A probability distribution is given by

$$\mathrm{P}(X = x) = \frac{x + 6}{40}$$

for $x = 1, 3, 5, 7$. Find $\mathrm{P}(X > 4)$

Question 4. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2	3	4 or more
$\mathrm{P}(X = x)$	0.05	0.1	k	0.35	0.32

Find the value of k

Question 5. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3	4	5
$\mathrm{P}(X = x)$	k	$3k$	$5k$	k	k

Find the value of k

Question 6. A probability distribution is given by

$$\mathrm{P}(X = x) = k(10 - x)$$

for $x = 2, 4, 6, 8$ where k is a constant.

a) Show that $k = \frac{1}{20}$

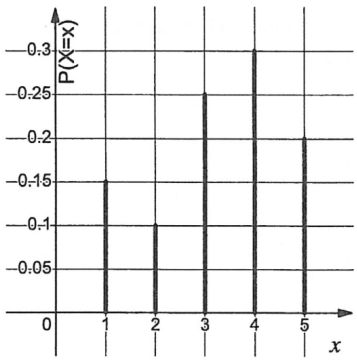
b) Calculate $\mathrm{P}(X \geq 6)$

Question 7. A probability distribution is given by

$$\mathrm{P}(X = x) = \frac{x + k}{10x + 10}$$

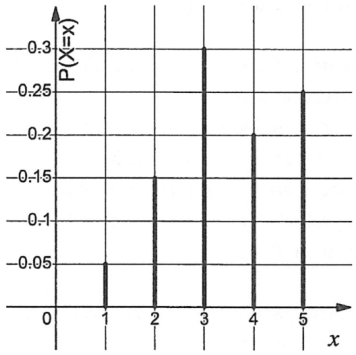
for $x = 1, 2, 3$. Find the value of k

Question 8. The stick graph below shows the probability distribution for the discrete random variable X.



Find $\mathrm{P}(1 < X \leq 4)$

Question 9. The stick graph below shows the probability distribution for the discrete random variable X.



Two values of X are selected at random. Assume these events are independent. Find the probability that both values are greater than or equal to 4

Question 10. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3
$\mathrm{P}(X = x)$	0.71	0.19	0.1

Two values of X are chosen at random. Assume these events are independent. Find the probability that both values are the same.

Question 11. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3
$\mathrm{P}(X = x)$	0.9	0.09	0.01

Two values of X are chosen at random. Assume these events are independent. Find the probability that the second value is less than the first.

Question 12. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3
$\mathrm{P}(X = x)$	0.24	0.36	0.4

Two values of X are chosen at random. Assume these events are independent. Find the probability that the first value is odd and the second value is prime.

Question 13. The table below shows the probability distribution for a discrete random variable X.

x	5	8	11
$\mathrm{P}(X = x)$	0.15	0.25	0.6

Two values of X are chosen at random. Assume these events are independent. Find the probability that the sum of the two scores is less than 17

Question 14. In a game, one spin of a biased spinner can score 1, 2, 3 or 4 points. The random variable X represents a player's score, and this has the following probability distribution:

x	1	2	3	4
$\mathrm{P}(X = x)$	0.02	0.14	0.26	0.58

The spinner is spun twice and the scores added. It is given that each game is independent. Find the probability of getting a total of 4 points.

Question 15. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2
$\mathrm{P}(X = x)$	0.33	0.22	0.45

Two values of X are chosen at random. Find the probability that the product of these two values is zero.

Question 16. The discrete random variable X is modelled by the probability distribution:

$$\mathrm{P}(X = x) = \frac{x}{20}$$

for $x = 3, 4, 6$ and 7. The independent random variables X_1 and X_2 have the same distribution as X. Find $\mathrm{P}(X_1 = X_2)$

Question 17. A spinner can be modelled by X with the probability distribution:

$$\mathrm{P}(X = x) = \frac{x+1}{14}$$

for $x = 1, 2, 3$ and 4. The spinner is spun three times. Find the probability that the first value is greater than the sum of the next two values.

Question 18. It is given that a discrete random variable X can take on the values 1, 2, 4 and 5, and that $\mathrm{P}(X = 1) = 0.2$. When $r = 1$ and $r = 2$, it is also given that $\mathrm{P}(X = r) = \mathrm{P}(X = r + 3)$
Draw a table to show the probability distribution of X.

Question 19. It is given that a discrete random variable X can take on the values 4, 5, 9 and 10, and that $\mathrm{P}(X = 5) = 0.18$. When $r = 4$ and $r = 5$, it is also given that $\mathrm{P}(X = r) = \mathrm{P}(X = r + 5)$
Draw a table to show the probability distribution of X.

Question 20. It is given that a discrete random variable X can take on the values 2, 3, 4, 5 and 6, and that $\mathrm{P}(X = 2) = 0.11$. When $r = 2$ and $r = 3$, it is also given that $\mathrm{P}(X = r) = \mathrm{P}(X = r + 3)$
It is also given that $\mathrm{P}(X = 4) = \mathrm{P}(X = 3) + \mathrm{P}(X = 5)$. Draw a table to show the probability distribution of X.

Question 21. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X = x) = \begin{cases} \frac{1}{5} \times \frac{2^{x-1}}{3^{x-1}} & x = 1, 2 \\ k & x = 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

a) Find $\mathrm{P}(X = 2)$.
b) Find the value of k
c) Find $\mathrm{P}(2 \leq X \leq 3)$

Question 22. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X = x) = \begin{cases} ax^2 & x = 1, 2 \\ bx + 0.05 & x = 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

where a and b are constants.

a) Find $\mathrm{P}(X = 2)$ in terms of a
b) Given that $\mathrm{P}(X = 2) = \mathrm{P}(X \geq 3)$, find the exact values of a and b

Question 23. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X = x) = \begin{cases} a(15 - 3x) & x = 1, 2, 3, 4 \\ bx & x = 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

where a and b are constants.

a) Show that $30a + 11b = 1$

b) Given that $\mathrm{P}(X \geq 4) = \frac{1}{4}$ find the exact values of a and b

Question 24. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2	3
$\mathrm{P}(X = x)$	$3k^2$	$3k$	$\frac{9}{2}k^2$	$\frac{1}{2}k$

Find the value of k and hence find $\mathrm{P}(X > 1)$

Question 25. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X=x)=\begin{cases}k & x=1,2,3,4\\ \frac{1}{3}\mathrm{P}(X=x-1) & x=5,6\\ 0 & \text{otherwise}\end{cases}$$

where k is a constant.
Show that $k=\frac{9}{40}$

Question 26. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X=x)=\begin{cases}\frac{1}{5}\mathrm{P}(X=x+1) & x=1,2,3\\ k & x=4,5\\ 0 & \text{otherwise}\end{cases}$$

where k is a constant. Find the value of k and hence find $\mathrm{P}(X\leq 3)$

Question 27. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2	3
$\mathrm{P}(X = x)$	$2k$	0.42	0.37	k

Two values of X are chosen at random. Find the probability that the product of these two values is greater than their sum.

Question 28. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2	3	4
$\mathrm{P}(X = x)$	$\frac{1}{9}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{5}{18}$

Two independent values, X_1 and X_2, of X are chosen. Find $\mathrm{P}(X_1 + X_2 = 5)$

Question 29. The table below shows the probability distribution for a discrete random variable X.

x	2	3	4	5
$\mathrm{P}(X = x)$	0.8	0.09	0.06	0.05

It is given that $Y = X^2$. Find $\mathrm{P}(Y + 2X \leq 15)$

Question 30. The table below shows the probability distribution for a discrete random variable X.

x	4	5	9	13
$\mathrm{P}(X = x)$	0.7	0.2	0.05	0.05

It is given that $Y = \frac{15}{X}$. Find $\mathrm{P}(X + Y \geq 8)$

Question 31. Tom models the number of games of pool he will win in a match by the random variable X with the following probability distribution:

x	0	1	2	3	More than 3
$\mathrm{P}(X = x)$	$\frac{1}{15}$	$\frac{1}{10}$	$\frac{1}{3}$	$\frac{1}{2}$	0

The number of games of pool won in a match is independent of any other match. Tom chooses three pool matches at random. Use the model to find the probability of:

a) Tom winning the same number of games of pool in all three matches;
b) Tom winning exactly two games of pool in total in the three matches;
c) Tom winning fewer games of pool than the preceding match across the three matches.

Question 32. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3	4
$\mathrm{P}(X = x)$	a	b	c	0.45

It is given that $\mathrm{P}(X \leq 2) = \frac{1}{3}\mathrm{P}(X \geq 3)$.
Find $\mathrm{P}(X \geq 3)$

Question 33. A probability distribution is given by

$$\mathrm{P}(X = x) = \begin{cases} \frac{k}{x+1} & x = 1, 3, 7 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant. Three independent values, X_1, X_2 and X_3, of X are chosen.
Find $\mathrm{P}(X_1 + X_2 > X_3)$

Question 34. The table below shows the probability distribution for a discrete random variable S.

s	6	12	18	24
$\mathrm{P}(S = s)$	$3k$	$4k$	$5k$	$6k$

S_1 and S_2 have the same distribution as S and are independent. Find $\mathrm{P}(S_1 + S_2 = 36)$

Question 35. Two bias spinners, X and Y, can be represented by the probability distributions given by $\mathrm{P}(X = x) = \frac{x}{25}$ for $x = 12, 13$, and $\mathrm{P}(Y = y) = \frac{y}{27}$ for $x = 13, 14$ respectively. Spinner X is spun first and then spinner Y is spun. Find the probability that the sum of the scores is 26

Question 36. The random variable X is a discrete uniform distribution where every outcome is equally likely. X can take on the values 1, 2, 3, 4 and 5. Two values, X_1 and X_2, of X are chosen at random. Find $\mathrm{P}(X_1 = X_2)$

Question 37. The random variable X is a discrete uniform distribution where every outcome is equally likely. X can take on the values $1, 2, 3, \ldots, n$. Two values, X_1 and X_2, of X are chosen at random. Find $\mathrm{P}(X_1 = X_2)$

Question 38. The random variable X is a discrete uniform distribution where every outcome is equally likely. X can take on the values $1, 2, 3, \ldots, n$. Two values, X_1 and X_2, of X are chosen at random. It is given that $\mathrm{P}(X_1 = X_2) = 0.1$. Find $\mathrm{P}(X_1 < X_2)$

Question 39. Prerequisite knowledge for Q39-Q40: Conditional Probability A probability distribution is given by

$$\mathrm{P}(X = x) = \begin{cases} 2kx & x = 3, 4, 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant. Two independent values, X_1 and X_2, of X are chosen.

a) Find $\mathrm{P}(X_1 + X_2 = 9)$

b) It is given that $X_1 + X_2 = 9$. Find the probability that one of X_1 or X_2 is 4

Question 40. A probability distribution is given by

$$\mathrm{P}(X = x) = \frac{2^x}{k}$$

for $x = 2, 3, 4, 5$ where k is a constant. Three independent values, X_1, X_2 and X_3, of X are chosen.

a) Find $\mathrm{P}(X_1 + X_2 + X_3 = 9)$

b) It is given that $X_1 + X_2 + X_3 = 9$. Find the probability that at least one of X_1, X_2 or X_3 is 4